1 Angles, Rotation, and the Unit Circle

This week's central question: How can we turn rotation into coordinates? The answer begins with the natural geometry of a circle.

The first half of precalculus introduced functions as ways of turning inputs into outputs. In this course, an angle is the input, and the outputs will be coordinates, distances, and measurements of motion. The geometry is worth understanding before trying to memorize a table of values. A circle gives us a concrete place to see what every formula is describing.

Learning goals. By the end of the chapter, you should be able to describe signed rotation, convert between degrees and radians, calculate arc length and angular speed, locate unit-circle points exactly, and explain how a turning point produces two repeating coordinate functions.

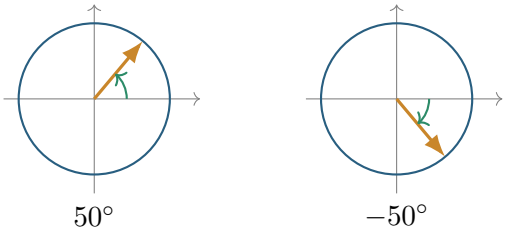
1.1 Angles measure turning

An angle records a rotation from one ray to another. Counterclockwise rotations are positive, and clockwise rotations are negative. A rotation can exceed one full revolution. For example, 450° and 90° have the same terminal ray.

Definition 1.1. Two angles are *coterminal* if they have the same initial and terminal rays. They differ by an integer multiple of 360°, or 2π radians.

Example 1.1. A positive coterminal angle for -50° is $-50^\circ + 360^\circ = 310^\circ$.

To make the meaning of a signed angle visible, imagine an arrow anchored at the origin. Its initial position points right. A positive angle turns the arrow counterclockwise; a negative angle turns it clockwise. The angle records the *turning*, not simply the final direction. Thus 30°, 390°, and -330° point along the same terminal ray, although their rotations are different.



Worked example: multiple turns. Find an angle between 0° and 360° coterminal with -770°. Adding 360° twice gives -50°; adding it once more gives 310°. This is the terminal direction even though the original rotation made more than two clockwise turns.

Try it. Sketch -120°, 240°, and 600°. Which terminal rays coincide? How can you tell without drawing all of the turns?

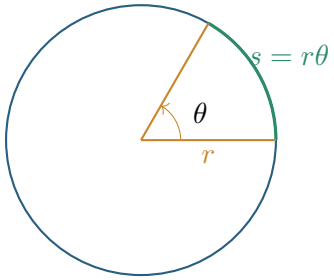
1.2 Why radians are natural

For a circle of radius r with intercepted arc length s, define the radian measure $\theta = s/r$. This is a ratio of lengths. A full circumference is $2\pi r$, so one revolution measures 2π radians:

$$\boxed{\theta = \frac{s}{r}}, \qquad 180^\circ = \pi \text{ radians}.$$

Example 1.2. Convert 150° to radians: $150(\pi/180) = 5\pi/6$. Convert $7\pi/4$ radians to degrees: $(7\pi/4)(180/\pi) = 315^\circ$.

Try it. Convert 225° to radians and $-\pi/3$ radians to degrees, and sketch their terminal rays.



Angle measured by the arc it intercepts

The radian definition has an important consequence: the radius cancels when we compare the corresponding arc length with the radius. A 60° turn subtends a longer arc on a large circle than on a small one, but the ratio s/r is the same. That is why the angle itself is independent of the circle's size.

Degrees	Radians	Fraction of one turn
30°	$\pi/6$	$1/12$
45°	$\pi/4$	$1/8$
60°	$\pi/3$	$1/6$
90°	$\pi/2$	$1/4$
180°	π	$1/2$
270°	$3\pi/2$	$3/4$
360°	2π	1

The table is not a second set of unrelated facts to memorize. Every entry comes from the single equality $180^\circ = \pi$ radians. For instance, 60° is one third of 180°, so it is $\pi/3$ radians.

Example 1.3. Convert a negative angle: $-210^\circ \cdot \pi/180^\circ = -7\pi/6$. Its terminal ray coincides with $5\pi/6$, because adding 2π to $-7\pi/6$ gives $5\pi/6$.

1.3 Arc length and angular speed

Rearrange the definition of radian measure to obtain $s = r\theta$. If rotation through θ takes time t, angular speed is $\omega = \theta/t$ and tangential speed is $v = r\omega$.

Example 1.4. A wheel of radius 0.4 m turns through $3\pi/2$ radians. A point on its rim travels $s = 0.4(3\pi/2) = 0.6\pi$ m.

Try it. A Ferris wheel of radius 12 m makes one revolution every 40 s. Find its angular speed and a passenger's tangential speed.

Worked example: a carousel. A carousel makes one revolution every 24 seconds. A child sits 3 meters from the axis, while another sits 5 meters from the axis. Both complete the same angular displacement in the same amount of time, so their angular speed is

$$\omega = \frac{2\pi}{24} = \frac{\pi}{12} \text{ rad/s}.$$

But their speeds along their respective circles differ:

$$v_1 = 3\frac{\pi}{12} = \frac{\pi}{4} \text{ m/s}, \qquad v_2 = 5\frac{\pi}{12} = \frac{5\pi}{12} \text{ m/s}.$$

The outer rider travels farther in the same time. This is the distinction between *angular speed* and *tangential speed*.

A caution about units. The formula $s = r\theta$ requires θ in radians. Substituting 90 for a 90° angle would calculate ninety radii of arc instead of a quarter-circumference. First convert 90° to $\pi/2$.

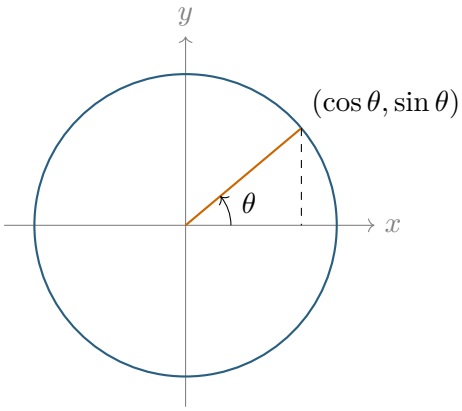
Try it. A wheel of radius 0.6 m turns through 135° in 3 seconds. Find the arc length, average angular speed, and average tangential speed. Explain the units of all three answers.

1.4 Rotation becomes coordinates

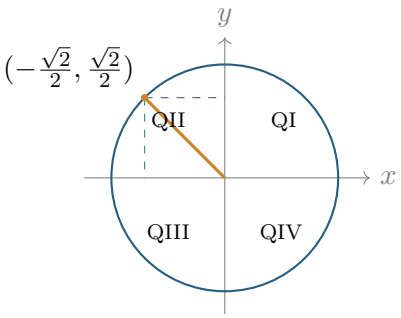
Take a circle of radius 1 centered at the origin. The point at angle θ, measured from the positive x-axis, has coordinates

$$\boxed{(x, y) = (\cos\theta, \sin\theta)}.$$

Cosine is the horizontal coordinate and sine is the vertical coordinate. Because the point lies on $x^2 + y^2 = 1$, we get $\cos^2\theta + \sin^2\theta = 1$.



We can now make the coordinate interpretation precise. Begin with the point $(1, 0)$ and rotate it through θ. The new point is $(\cos\theta, \sin\theta)$. The horizontal coordinate can be negative, and so can the vertical coordinate: the point is not confined to a triangle in the first quadrant.



Worked example: a point in Quadrant III. At $225° = 5\pi/4$, the reference angle is $\pi/4$. The two coordinates have equal absolute value, but both are negative:

$$(\cos(5\pi/4), \sin(5\pi/4)) = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right).$$

This example illustrates why taking the positive side lengths of a reference triangle is only the first step. The signs come from the location in the plane.

Try it. Without a calculator, describe the point on the unit circle at 150° and at 330°. Which coordinates have the same absolute value? Which signs differ?

1.5 Reference angles and periodicity

Special angles arise from right triangles:

θ	$\cos\theta$	$\sin\theta$
0	1	0
$\pi/6$	$\sqrt{3}/2$	$1/2$
$\pi/4$	$\sqrt{2}/2$	$\sqrt{2}/2$
$\pi/3$	$1/2$	$\sqrt{3}/2$
$\pi/2$	0	1

The reference angle supplies the absolute values in every quadrant; the quadrant gives the signs.

Example 1.5. At $5\pi/6$, the reference angle is $\pi/6$ and the point is in quadrant II, so $\cos(5\pi/6) = -\sqrt{3}/2$ and $\sin(5\pi/6) = 1/2$.

After one full revolution the point returns to its previous position:

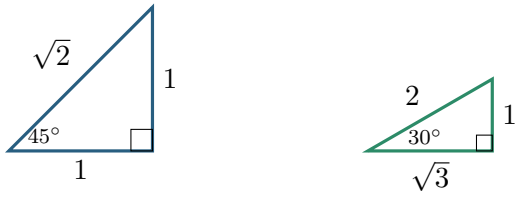
$$\sin(\theta + 2\pi) = \sin\theta, \qquad \cos(\theta + 2\pi) = \cos\theta.$$

This geometric periodicity will become the foundation of sinusoidal graphs.

1.6 Where exact coordinates come from

Instead of treating the exact-value table as a memorization task, we can derive its entries from two familiar triangles.

An isosceles right triangle with legs 1 has hypotenuse $\sqrt{2}$. Dividing both legs by $\sqrt{2}$ yields the point $(\sqrt{2}/2, \sqrt{2}/2)$ at 45°. Likewise, a 30°–60°–90° triangle has side lengths in the ratio $1 : \sqrt{3} : 2$. Dividing by its hypotenuse produces the coordinates $(\sqrt{3}/2, 1/2)$ for 30° and $(1/2, \sqrt{3}/2)$ for 60°.



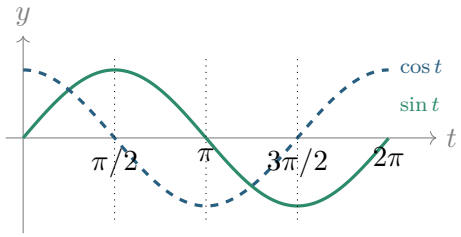
The picture is a reminder that exact values are ratios of lengths: the unit circle merely normalizes the hypotenuse to 1.

1.7 From the circle to a wave

There is one more idea that connects this chapter to the rest of the course. Suppose the angle increases steadily in time:

$$\theta(t) = \omega t, \qquad x(t) = \cos(\omega t), \quad y(t) = \sin(\omega t).$$

The moving point goes around a circle, but the horizontal and vertical coordinates each rise and fall repeatedly. Plotting either coordinate against time produces a wave.



This is a picture of a *coordinate varying over time*, not the circular path itself. The distinction will matter when we study graphs of periodic functions in Module 3.

1.8 A complete mathematical model

A Ferris wheel has radius 8 meters, its center is 10 meters above ground, and a passenger starts at the rightmost point. Suppose it rotates counterclockwise once every 40 seconds. Its angular speed is $\omega = 2\pi/40 = \pi/20$ radians per second, and the passenger's position relative to the ground is

$$\boxed{x(t) = 8\cos\left(\frac{\pi t}{20}\right), \qquad y(t) = 10 + 8\sin\left(\frac{\pi t}{20}\right).}$$

At $t = 0$ the passenger is at $(8, 10)$. At $t = 10$ seconds the wheel has turned through $\pi/2$, so the passenger is at $(0, 18)$. At $t = 20$ seconds the passenger is at $(-8, 10)$. The model answers position questions without tracking the entire path by hand.

Interpreting the constants. The radius appears as the amplitude of each coordinate function. The center height appears as a vertical translation. The revolution time appears in the angular speed. Each physical fact has an identifiable place in the equations.

Try it. A different wheel has radius 6 meters, center height 7 meters, and period 30 seconds. Starting from the rightmost point, write its coordinate model and find the passenger's height after 7.5 seconds.

1.9 Exercises

1. Convert $-135°$ and $300°$ to radians.
2. Convert $11\pi/6$ and $-5\pi/4$ to degrees.
3. Find an angle in $[0, 2\pi)$ coterminal with $19\pi/6$.
4. A circle of radius 8 cm subtends an angle of $5\pi/12$. Find the arc length.
5. A turntable makes $33\frac{1}{3}$ revolutions per minute. Find angular speed in radians per second.
6. Explain why sine and cosine must lie between -1 and 1.
7. Find unit-circle coordinates at $2\pi/3$, $5\pi/4$, and $11\pi/6$.
8. A Ferris wheel has radius 10 m and center $(0, 12)$. Write a rider's coordinates after turning through θ from the rightmost point.
9. A point begins at $(1, 0)$ and turns through $-7\pi/6$ radians. Determine a coterminal angle in $[0, 2\pi)$ and give the exact point.
10. A wheel completes 9 revolutions in 2 minutes. Determine its angular speed in both radians per second and degrees per second.

11. A rider on a circle of radius 4 m moves through 75°. Find the arc length exactly in terms of π and approximately to two decimals.

12. Describe the coordinates at $3\pi/2$ and explain why a terminal point on an axis does not belong to any open quadrant.

13. For the Ferris-wheel model in the chapter, explain which physical quantity would change if the coefficient 8 in both coordinate equations became 12.

Looking ahead. Next week we extend sine and cosine to the remaining four trigonometric functions.